

Student: Jonatan Rabjerg
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Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 13

Find all the local maxima, local minima, and saddle points of the function.

$$f(x,y) = 7e^{-y}(x^2 + y^2) + 8$$

The local extrema and saddle points in the interior of the domain of a function occur at critical points of the function. An interior point of the domain is a critical point if both f_x and f_y are zero or if one or both derivatives do not exist.

Find the partial derivative f_x .

$$f(x,y) = 7e^{-y}(x^2 + y^2) + 8$$

$$f(x,y) = 7e^{-y}x^2 + 7e^{-y}y^2 + 8$$

$$f_x = \frac{14x}{e^y}$$

Expand.

Differentiate. Note that the last two terms don't involve x .

Find the partial derivative f_y .

$$f(x,y) = 7e^{-y}(x^2 + y^2) + 8$$

$$f_y = 7(-e^{-y}(x^2 + y^2) + e^{-y}(2y))$$

Apply the Product Rule.

$$f_y = \frac{14y - 7(x^2 + y^2)}{e^y}$$

Simplify.

Both derivatives exist everywhere in the domain of $f(x,y)$. Since the critical points exist where both f_x and f_y are zero, solve $f_x = 0$ for x .

$$\frac{14x}{e^y} = 0$$

$$x = 0$$

To find the corresponding y -value(s), substitute the resulting value of x into f_y and solve the equation $f_y = 0$ for y .

$$\frac{14y - 7(x^2 + y^2)}{e^y} = 0$$

$$\frac{14y - 7((0)^2 + y^2)}{e^y} = 0$$

$$\frac{14y - 7y^2}{e^y} = 0$$

Solve for y .

$$\frac{14y - 7y^2}{e^y} = 0$$

$$14y - 7y^2 = 0$$

$$7y(2 - y) = 0$$

$$y = 0, 2$$

Therefore, the critical points are $(0,0)$ and $(0,2)$.

The nature of the critical points can be tested using the second derivative test if the first- and second-order partial derivatives of f are continuous throughout a disk centered at point (a,b) and $f_x(a,b) = f_y(a,b) = 0$. The second derivative test is stated below.

i) f has a local maximum at (a,b) if $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a,b) .

ii) f has a local minimum at (a,b) if $f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a,b) .

iii) f has a saddle point at (a,b) if $f_{xx}f_{yy} - f_{xy}^2 < 0$ at (a,b) .

iv) The test is inconclusive at (a,b) if $f_{xx}f_{yy} - f_{xy}^2 = 0$ at (a,b) .

The expression $f_{xx}f_{yy} - f_{xy}^2$ is called the discriminant. First find f_{xx} by differentiating f_x with respect to x .

$$f_x = \frac{14x}{e^y}$$

$$f_{xx} = \frac{14}{e^y}$$

Differentiate f_y with respect to y to find f_{yy} .

$$f_y = \frac{14y - 7(x^2 + y^2)}{e^y}$$

$$f_{yy} = \frac{e^y(14 - 0 - 14y) - [14y - 7(x^2 + y^2)]e^y}{e^{2y}} \quad \text{Use the quotient rule.}$$

$$f_{yy} = \frac{14 - 28y + 7x^2 + 7y^2}{e^y} \quad \text{Simplify.}$$

Differentiate f_x with respect to y to find f_{xy} .

$$f_x = \frac{14x}{e^y}$$

$$f_{xy} = -\frac{14x}{e^y}$$

Evaluate the discriminant at $(0,0)$ to determine the nature of the first critical point.

$$\begin{aligned} f_{xx}f_{yy} - f_{xy}^2 &= \left(\frac{14}{e^y}\right)\left(\frac{14 - 28y + 7x^2 + 7y^2}{e^y}\right) - \left(-\frac{14x}{e^y}\right)^2 \\ &= \left(\frac{14}{e^{(0)}}\right)\left(\frac{14 - 28(0) + 7(0)^2 + 7(0)^2}{e^{(0)}}\right) - \left(-\frac{14(0)}{e^{(0)}}\right)^2 \\ &= 196 \end{aligned}$$

At this point $(0,0)$, the discriminant is 196 and f_{xx} is 14. Thus, the first critical point is a local minimum.

Evaluate the discriminant at $(0,2)$ to determine the nature of the second critical point.

$$\begin{aligned} f_{xx}f_{yy} - f_{xy}^2 &= \left(\frac{14}{e^y}\right)\left(\frac{14 - 28y + 7x^2 + 7y^2}{e^y}\right) - \left(-\frac{14x}{e^y}\right)^2 \\ &= \left(\frac{14}{e^{(2)}}\right)\left(\frac{14 - 28(2) + 7(0)^2 + 7(2)^2}{e^{(2)}}\right) - \left(-\frac{14(0)}{e^{(2)}}\right)^2 \\ &= -\frac{196}{e^4} \end{aligned}$$

At the point $(0,2)$, the discriminant is $-\frac{196}{e^4}$ and f_{xx} is $\frac{14}{e^2}$. Thus, the second critical point is a saddle point.

Recall that there is a local minimum at $(0,0)$. Find the value of the function at that point by substituting the values of the coordinates into the function.

$$\begin{aligned} f(x,y) &= 7e^{-y}(x^2 + y^2) + 8 \\ f(0,0) &= 7e^{(0)}((0)^2 + (0)^2) + 8 \\ &= 8 \end{aligned}$$

Therefore, the function has a minimum value of 8 at $(0,0)$, there is a saddle point at $(0,2)$, and the function has no local maximum.